2. ReactJS-HOL

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1. Explain React Components

A **React component** is an independent and reusable piece of code that defines a part of a user interface (UI). Components help divide the UI into smaller, manageable pieces, making applications easier to develop and maintain.

Features:

- Reusable
- Independent
- Accepts input through **props**
- Can manage data using **state**
- Returns JSX to display UI

Example:

```
function Welcome() {
  return <h1>Welcome to React!</h1>;
}
```

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2. Identify the Differences Between Components and JavaScript Functions

React Component	JavaScript Function
Used to create UI elements.	Used to perform general programming tasks.
Returns JSX (HTML-like syntax).	Returns values such as numbers, strings, or objects.
Can receive **props** and manage **state**.	Receives parameters but does not manage React state.
Used within React applications.	Can be used in any JavaScript program.
Component names must start with a capital letter.	Function names can start with lowercase or uppercase.

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3. Identify the Types of Components

There are **two main types of React components**:

1. Function Components

- Written as JavaScript functions.
- Simpler and easier to write.
- Can use Hooks (e.g., useState, useEffect) for state and lifecycle features.

Example:

```
function Home() {
  return <h1>Home Page</h1>;
}
```

2. Class Components

- Written using ES6 classes.
- Extend React.Component.
- Use render() to display UI.
- Can manage state using this.state.

Example:

```
class Home extends React.Component {
  render() {
    return <h1>Home Page</h1>;
  }
}
```

---

4. Explain Class Component

A **class component** is a React component created using an ES6 class that extends React.Component. It must include a render() method that returns JSX.

Features:

- Uses this.state for state management.
- Can use lifecycle methods.
- Must extend React.Component.

- Uses render() to display UI.

Example:

```
import React, { Component } from "react";

class Home extends Component {
  render() {
    return <h1>Welcome to Home</h1>;
  }
}

export default Home;
```

---

5. Explain Function Component

A **function component** is a JavaScript function that returns JSX. It is the modern and preferred way of creating React components.

Features:

- Simple and lightweight.
- Easy to read and maintain.
- Uses Hooks for state and lifecycle management.
- Does not require render().

Example:

```
function Home() {
  return <h1>Welcome to Home</h1>;
}

export default Home;
```

---

6. Define Component Constructor

A **constructor** is a special method in a class component that is called when the component is created. It is mainly used to initialize the component's state and bind event handlers.

Syntax:

```
constructor(props) {
  super(props);
  this.state = {
    name: "Palak"
  };
}
```

Purpose:

- Initializes state.
- Receives props.
- Binds methods (if required).

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7. Define render() Function

The **render()** function is a mandatory method in every React class component. It returns the JSX that should be displayed on the screen.

Features:

- Returns JSX or React elements.
- Called whenever the component's state or props change.
- Responsible for rendering the UI.

Example:

```
class Home extends React.Component {
  render() {
    return (
      <h1>Welcome to React</h1>
    );
  }
}
```